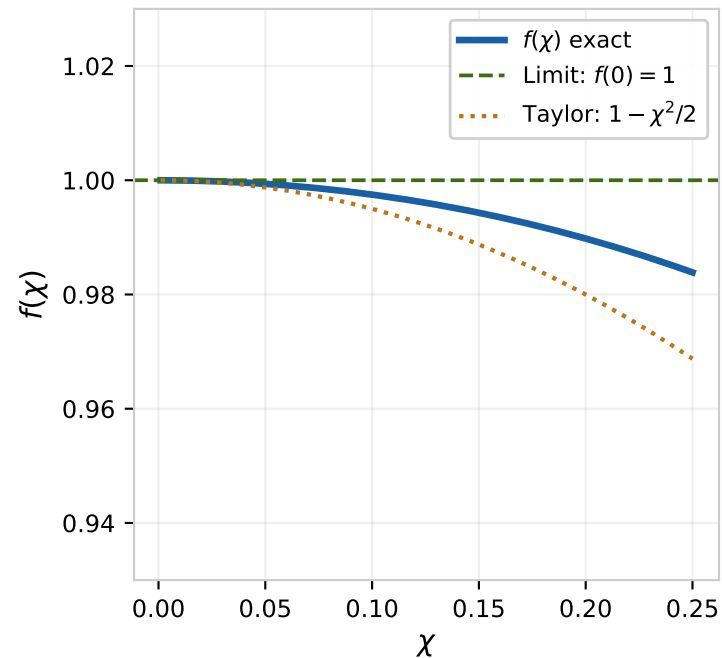
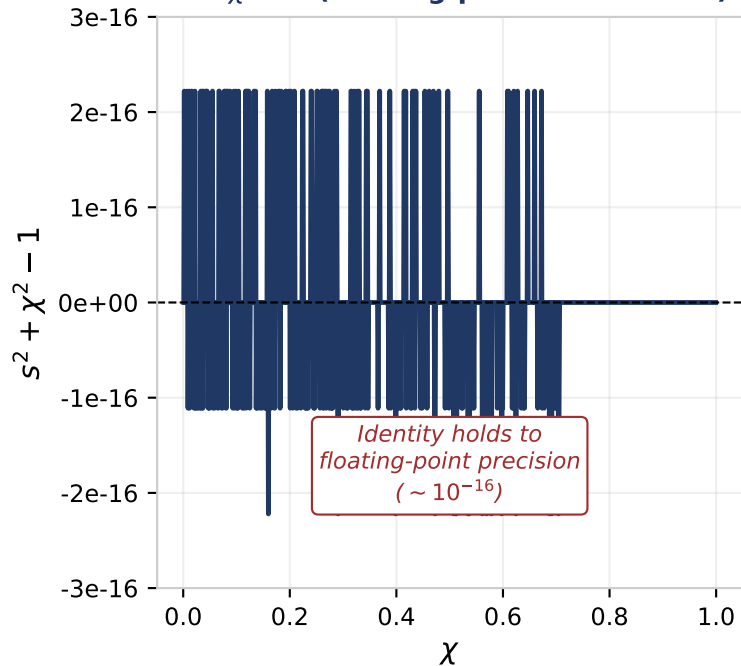


Verification of exact limits and the key algebraic identity

(a) Schwarzschild limit
 $f(0) = 1$ exactly



(b) Key identity
 $s^2 + \chi^2 = 1$ (floating-point verification)



(c) Extremal limit
 $f \rightarrow 0$ as $\chi \rightarrow 1$, approx $2\sqrt{2\epsilon}$

